

HW 8

review questions and exercises

1. The indirection operator dereferences a pointer, it allows you to access the item that the pointer points to.
7. New operator is used to allocate memory. New returns address of memory location.
9. Delete operator is used to free dynamic memory
11. Pointer to a constant: you cannot modify the value that pointer points to through the pointer. Constant pointer: You can modify the value that the pointer points to through the pointer (the pointer's address can't)
13. address
14. address
21. null
24. `cout << * ptr << endl;`
32. T
41. F
48. delete the * before nullptr
51. `ptr = 100;` should be `* ptr = 100;`

programming challenges

```
1  #include <iostream>
2  using namespace std;
3
4  int maxtimes(int *arr, int size)
5  {
6      int count = 0;
7      for (int i = 0; i < size; i++)
8      {
9          int presentvalue = *(arr + i);
10         int tempcount = 0;
11         for (int j = 0; j < size; j++)
12         {
13             if (*(arr + j) == presentvalue)
14             {
15                 tempcount++;
16             }
17         }
18         if (tempcount > count)
19         {
20             count = tempcount;
21         }
22     }
23     if (count == 1)
24     {
25         count = -1;
26     }
27     return count;
28 }
29
30 int main()
31 {
32     const int SIZE = 10;
33     int arr[SIZE] = {1, 2, 3, 4, 5, 3, 2, 1, 3, 4};
34     int result = maxtimes(arr, SIZE);
35     cout << result << endl;
36     return 0;
37 }
```

3

Process exited after 0.1969 seconds with return value 0
请按任意键继续. . .